Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The area of the figure bounded by $y^2 = 9x$ and $y = 3x$ is

(A) 1

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 2

Q.2

Let $f(x)$ be a continuous function such that the area bounded by the curve $y = f(x)$, the x-axis, and the lines $x = 0$ and $x = a$ is $1 + \frac{a^2}{2}\sin a$. Then

(A) $f\left(\frac{\pi}{2}\right) = 1 + \frac{\pi^2}{8}$

(B) $f(a) = 1 + \frac{a^2}{2}\sin a$

(C) $f(a) = a\sin a + \frac{1}{2}a^2\cos a$

(D) none of these

Q.3

Area bounded by the curve $y = |x| - 2$ and $y = 1 - |x - 1|$ is equal to

(A) 4 sq. units

(B) 6 sq. units

(C) 2 sq. units

(D) 8 sq. units

Q.4

Area bounded by $y = \tan^{-1} x, y = \cot^{-1} x$ and y-axis is equal to

(A) In $\sqrt{2}$ sq. units

(B) $\ln 4$ sq. units

(C) $\ln 8$ sq. units

(D) $\ln 2$ sq. units

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Q.5

Area of the triangle formed by the x-axis and tangent and normal drawn to the curve $y = x^2$ at $(1, 1)$ is equal to

(A) 3 sq. units

(B) 5/4 sq. units

(C) 4 sq. units

(D) 2 sq. units

Q.6

Area enclosed by the curve $y = f(x)$ that is being defined parametrically as $x = \frac{1-t^2}{1+t^2}, y = \frac{2t}{1+t^2}$ is equal to

(A) π sq. units

(B) $\frac{\pi}{2}$ sq. units

(C) $\frac{3\pi}{4}$ sq. units

(D) $\frac{3\pi}{2}$ sq. units

Q.7

Area of the region bounded by the curves $y = 2^x, y = 2x - x^2, x = 0$ and $x = 2$ is given by

(A) $\frac{3}{\log 2} - \frac{4}{3}$

(B) $\frac{3}{\log 2} + \frac{4}{3}$

(C) $3 \log 2 - \frac{4}{3}$

(D) none of these

Q.8

The area of the figure bounded by $y = \sin x, y = \cos x$ in the first quadrant is

(A) $\sqrt{2}(\sqrt{2} - 1)$

(B) $\sqrt{3} + 1$

(C) $2(\sqrt{3} - 1)$

(D) none of these

Q.9

The area bounded by the curve $|x| + |y| \geq 1$ and $x^2 + y^2 \leq 1$ is

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(A) 2 sq. units

(B) π sq. units

(C) $(\pi - 2)$ sq. units

(D) $(\pi + 2)$ sq. units

Q.10

The area bounded by $y = \frac{\sin x}{x}$, x-axis and the ordinates $x = 0, x = \frac{\pi}{4}$ is Δ, then which of the following is not true,

(A) Δ is less than $\frac{\pi}{2}$

(B) Δ is less than $\frac{\pi}{4}$

(C) Δ is greater than $\frac{\pi}{4}$ than $\int_0^{\pi/4} \frac{\tan x}{x} dx$

(D) Δ is less than $\frac{3\pi}{4}$

Q.11

The area bounded by $y = 2 - |2 - x|$ and $y = \frac{3}{|x|}$ is

(A) $\frac{4+3\log 3}{2}$ sq. units

(B) $\frac{4-3\log 3}{2}$ sq. units

(C) $\frac{3}{2}\log 3$ sq. units

(D) $\frac{1}{2} + \log 3$ sq. units

Q.12

Area bounded by the curves $y = \sin \frac{\pi x}{2}$ and $y = x^3$, is equal to

(A) $\frac{4-\pi}{\pi}$ sq. units

(B) $\frac{4-\pi}{2\pi}$ sq. units

(C) $\frac{8-\pi}{\pi}$ sq. units

(D) $\frac{8-\pi}{2\pi}$ sq. Units

Q.13

Area enclosed by the curve $\left(y - \sin^{-1} x\right)^2 = x - x^2$, is equal to

(A) $\frac{\pi}{2}$ sq. units

(B) $\frac{\pi}{4}$ sq. units

(C) $\frac{\pi}{8}$ sq. units

(D) none of these

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Q.14

The area common to the parabola $y = 2x^2$ and $y = x^2 + 4$ is

(A) $\frac{2}{3}$ sq. units

(B) $\frac{3}{2}$ sq. units

(C) $\frac{32}{3}$ sq. units

(D) $\frac{3}{32}$ sq.units

Q.15

The area bounded by $\min(|x|, |y|) = 2$ and $\max(|x|, |y|) = 4$ is

(A) 4 sq. unit

(B) 16 sq. unit

(C) 8 sq. unit

(D) 12 sq. unit

Q.16

If a curve $y = a\sqrt{x} + bx$ passes through the point $(1, 2)$ and the area bounded by the curve, line $x = 4$ and x-axis is 8 sq. unit, then

(A) $a = 3, b = -1$

(B) $a = 3, b = 1$

(C) $a = -3, b = 1$

(D) $a = -3, b = -1$

Q.17

The area bounded by the curve $xy - 3x - 2y - 10 = 0$, x-axis and straight lines $x = 3, x = 4$ is

(A) $16 \log 2 - 13$

(B) $16 \log 2 - 3$

(C) $16 \log 2 + 3$

(D) none of these

Q.18

The area bounded by the curve $y = \ln x$, the x-axis and the line $x = e$ is

(A) e sq. units

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(B) 1 sq. unit

(C) $\left(1-\frac{1}{e}\right)$ sq. units

(D) $\left(1+\frac{1}{e}\right)$ sq. units

Q.19

The area between the hyperbola $xy = c^2$, then x-axes and the ordinates at a and b with $0 < a < b$ is

(A) $c^2 \log\left(\frac{a}{b}\right)$

(B) $c^2 \log\left(\frac{b}{a}\right)$

(C) $c^2 \log(ab)$

(D) none of these

Q.20

The slope of the tangent of the curve $y = f(x)$ at $\{x, f(x)\}$ is $(2x+1)$. If the curve passes through the point $(1, 2)$, then area bounded by the curve, x-axis and the line $x = 1$ is

(A) $\frac{5}{6}$

(B) $\frac{6}{5}$

(C) $\frac{1}{6}$

(D) 6

Q.21

Area of region satisfying $x \leq 2, y \leq |x|$ and $y \geq 0$ is

Q.22

The area of the region bounded by $y = |x-1|$ and $y = 1$ is

Q.23

Area lying in the first quadrant and bounded by the circle $x^2 + y^2 = 4$, the line $x = \sqrt{3}y$ and x-axis is $\frac{\pi}{n}$ where $n =$

Q.24

Area bounded by the curve $y = |x| - 2$ and $y = 1 - |x-1|$ is equal to

Q.25

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The area bounded by the curve $y = 2\cos x$ and the x-axis from $x = 0$ to $x = 2\pi$, is (in sq. units)

Q.26

Let $\int_0^{e^x} \left(t^2 - 1\right) f(t) = \frac{1}{15}\left[3e^{5x} - 5e^{3x} + 15\right]$. Then find the area rounded off to the nearest integer bounded by $y = f(x)$ and $y = 4$.

Q.27

The area bounded by the curves $y = \ln x, y = \ln|x|, y = |\ln x|$ and $y = |\ln|x||$ is

Q.28

Area bounded by $y = f(x) = \min(x + 1, \sqrt{1 - x})$ for all $x \leq 1$ is m/n where m and n are coprime then $m.n =$

Q.29

Let $F(x)$ be the area bounded by the curve $f(t) = \frac{e^t}{t}$ between $t = a(a > 1), t = x$ and axis of abscissa then the area bounded by $g(t) = \frac{e^t}{t+t_0}\ (t_0 > 0)$ between $t = a, t = x$ and axis of abscissa is given by $e^{-pt_0}\left[qF\left(x + t_0\right) + rF\left(a + t_0\right)\right]$ where p, q, r are natural numbers. The value of $(p + q + r)! =$

Q.30

The value of $b^3(b > 0)$ for which the area bounded by the curves $y = \frac{x}{6} + \frac{1}{x^2}$, x-axis, $x = b$ and $x = 2b$ has the least value is

Answers & Solutions

Q.1 Answer:

$\frac{1}{2}$

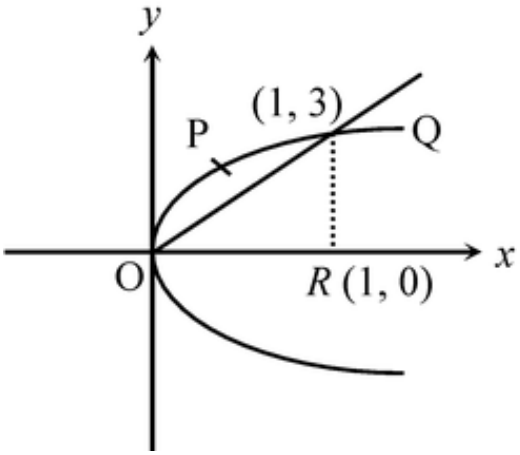
Solution:

Required area $=$ **area** $OPQRO -$ **area** ΔOQR

$\int_0^1 \sqrt{9x}dx - \frac{1}{2} \times 1 \times 3$

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$= 3\frac{2}{3}x^{3/2}\Big|_0^1 - \frac{3}{2} = \frac{1}{2}$



Q.2 Answer:

$f(a) = a\sin a + \frac{1}{2}a^2\cos a$

Solution:

Here $\int_0^a f(x)dx = 1 + \frac{a^2}{2}\sin a.$

Differentiating w.r.t. a, $f(a) = a\sin a + \frac{a^2}{2}\cos a$

Q.3 Answer:

4 sq. units

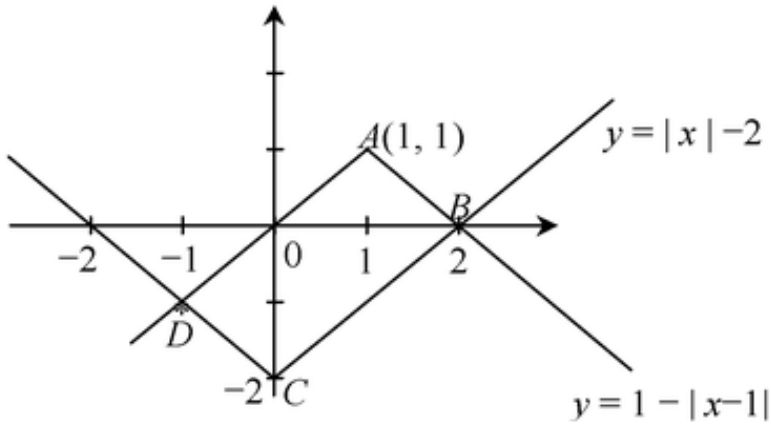
Solution:

Bounded figure $ABCD$ **is a rectangle.**

$AB = \sqrt{1+1} = \sqrt{2}$

$BC = \sqrt{4+4} = 2\sqrt{2}$

Thus, bounded area $= (\sqrt{2})(2\sqrt{2}) = 4$ **sq. units.**



Q.4 Answer:

$\ln 2$ sq. units

Solution:

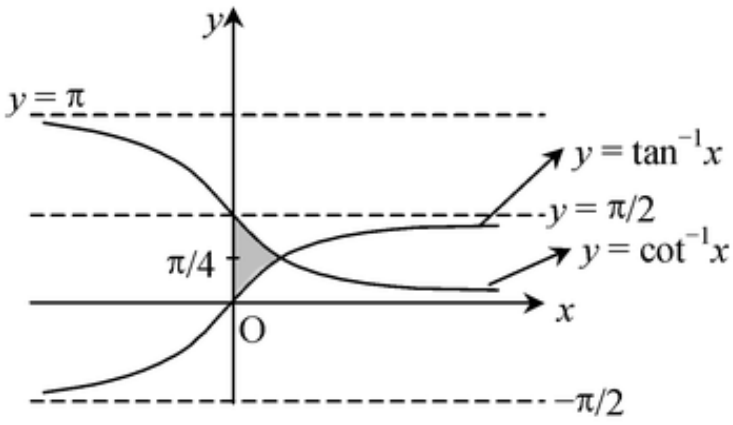
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Required area,

$$\Delta = \int_0^{\pi/4} \tan y dy + \int_{\pi/4}^{\pi/2} \cot y dy$$

$$= \ln \sec y|_0^{\pi/4} + \ln \sin y|_{5\pi/4}^{\pi/2}$$

$$= \ln \sqrt{2} - \ln \frac{1}{\sqrt{2}} = \ln 2$$

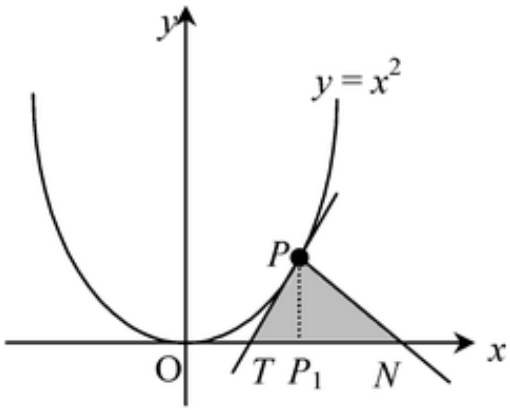


Q.5 Answer:

5/4 sq. units

Solution:

$y = x^2 \Rightarrow \frac{dy}{dx} = 2x$



Equation of tangent at $P(1,1)$ is $(y-1) = 2(x-1)$ i.e. $y = 2x - 1$. Thus $T \equiv \left(\frac{1}{2}, 0\right)$.
Equation of normal PN at $P(1,1)$ is, $(y-1) = -\frac{1}{2}(x-1)$ i.e. $2y = -x + 3$. Thus N=(3,0)
Required area $= \Delta_{PP_1T} + \Delta_{PP_1N} = \frac{1}{2} \cdot 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot 2 \cdot 1 = \frac{5}{4}$ sq. units.

Q.6 Answer:

π sq. units

Solution:

Clearly t can be any real number.

Let $t = \tan\theta \Rightarrow \quad x = \frac{1-\tan^2\theta}{1+\tan^2\theta} \Rightarrow x = \cos 2\theta$

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$y = \frac{2\tan\theta}{1+\tan^2\theta} = \sin 2\theta$

and $\Rightarrow x^2 + y^2 = 1$

Thus required area $= \pi.1^2 = \pi$ **sq. units.**

Q.7 Answer:

$\frac{3}{\log 2} - \frac{4}{3}$

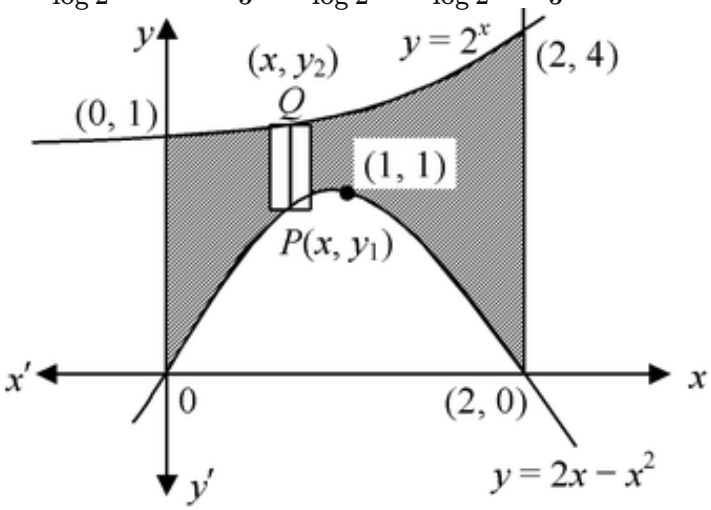
Solution:

Required area

$= \int_0^2 (y_2 - y_1)\, dx \int_0^2 \left(2^x - \left(2x - x^2\right)\right) dx$

$= \left[\frac{2^x}{\log 2} - x^2 + \frac{x^3}{3}\right]_0^2$

$= \frac{4}{\log 2} - 4 + \frac{8}{3} - \frac{1}{\log 2} = \frac{3}{\log 2} - \frac{4}{3}$



Q.8 Answer:

$\sqrt{2}(\sqrt{2} - 1)$

Solution:

Curve $y = \sin x$ **and** $y = \cos x$ **intersect at point** A**.**

$\therefore \sin x = \cos x$

$\Rightarrow x = \frac{\pi}{4}$

Required area = Area OAC + Area ACB

$= \int_0^{\pi/4} y_1 dx + \int_{\pi/2}^{\pi/2} y_2 dx$

$= \int_0^{\pi/4} \sin x dx + \int_{\pi/4}^{\pi/2} \cos x dx$

$= [-\cos x]_0^{\pi/4} + [\sin x]_{\pi/4}^{\pi/2}$

$= \left(-\frac{1}{\sqrt{2}} + 1\right) + \left(1 - \frac{1}{\sqrt{2}}\right)$

$= 2 - \frac{2}{\sqrt{2}} = \frac{2}{\sqrt{2}}(\sqrt{2} - 1) = \sqrt{2}(\sqrt{2} - 1)$

Q.9 Answer:

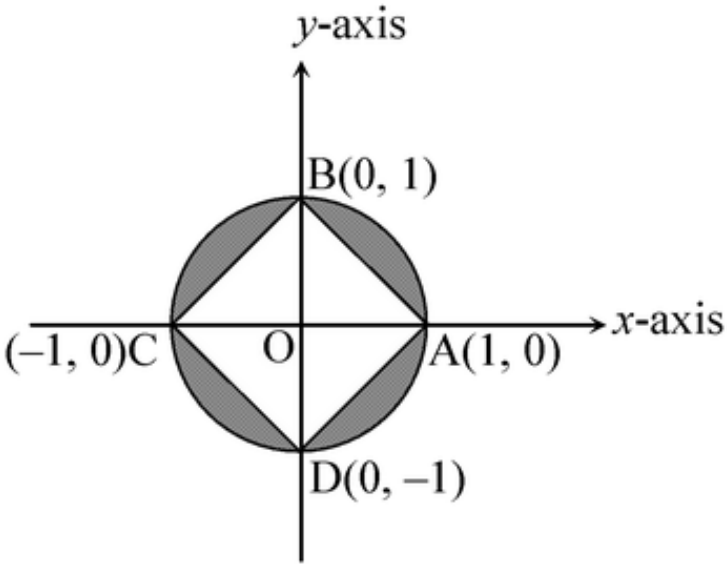
$(\pi - 2)$ sq. units

Solution:

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Area of square $ABCD = 2$ sq. units area of circle $= \pi$ sq. units

$\Rightarrow$ Required area $= (\pi - 2)$ sq. units



Q.10 Answer:

Δ is greater than $\frac{\pi}{4}$ than $\int_0^{\pi/4} \frac{\tan x}{x} dx$

Solution:

Since $\frac{\sin x}{x} \leq 1 \quad \Rightarrow \int_0^{\pi/4} \frac{\sin x}{x} dx \leq \int_0^{\pi/4} 1 dx \quad \Rightarrow \int_0^{\pi/4} \frac{\sin x}{x} dx \leq \frac{\pi}{4}$

Also $\int_0^{\pi/4} dx \leq \int_0^{\pi/4} \frac{\tan x}{x} dx$ (in the similar way).

Q.11 Answer:

$\frac{4-3\log 3}{2}$ sq. units

Solution:

$y = 2 - |2 - x| y = \frac{3}{|x|}$

1 can be rewritten as $y = xi f x < +2$

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On solving these two equation, we get

$x = \sqrt{3}$ and $x = 3$

$$\text{Area } = \int_{\sqrt{3}}^{3} (y_1 - y_2)\, dx$$

$$= \int_{\sqrt{3}}^{3} (2 - |2 - x|) - \frac{3}{|x|} dx$$

$$= \int_{\sqrt{3}}^{2} x - \frac{3}{|x|} + \int_{2}^{3} 4 - x - \frac{3}{|x|} dx$$

$$\left[\frac{x^2}{2} - 3\log_e x\right]_{\sqrt{3}}^{2} + \left[4x - \frac{x^2}{2} - 3\log_e x\right]_{2}^{3}$$

$$= \frac{1}{2} + 3\log_e \frac{\sqrt{3}}{2} + 12 - \frac{9}{2} - 3\log_e 3 - 8 + 2 + 3\log_e 2$$

$$= \frac{1}{2} + 3\ln \frac{\sqrt{3}}{2} + \frac{3}{2} 3\ln \frac{2}{3}$$

$$2 + 3\ln \frac{\sqrt{3}}{2} = 2 - 3\ln \sqrt{3} = \frac{4 - 3\ln 3}{2}$$

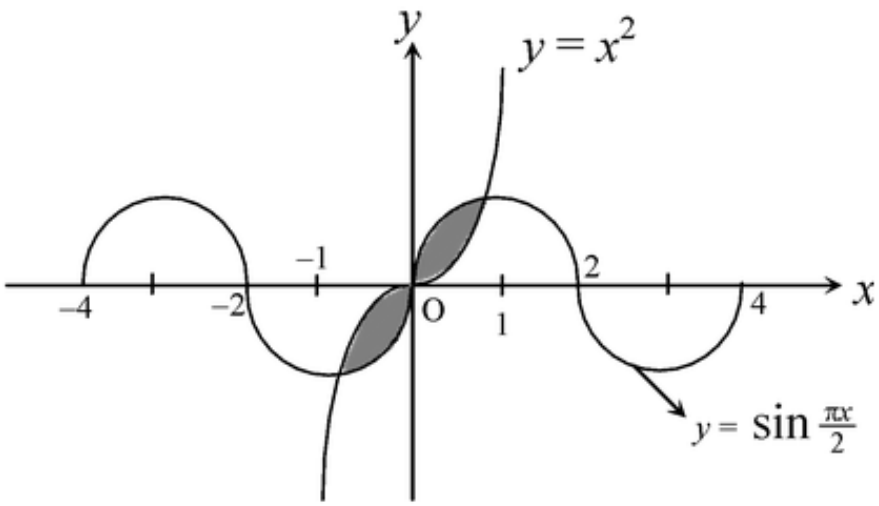
Q.12 Answer:

$\frac{8-\pi}{2\pi}$ sq. Units

Solution:

Bounded area $\Delta = 2\int_0^1 \left(\sin \frac{\pi x}{2} - x^3\right) dx$

$= 2\left(-\frac{2}{\pi}\cos \frac{\pi x}{2} - \frac{x^4}{4}\right)_0^1 = 2\left(\frac{2}{\pi} - \frac{1}{4}\right) = \frac{8-\pi}{2\pi}$



Q.13 Answer:

$\frac{\pi}{4}$ sq. units

Solution:

Clearly $x \leq 1$ **and** $x - x^2 \geq 0 \Rightarrow x \in [0, 1]$.

Curve has two branches whose equations are $y = \sin^{-1} x \pm \sqrt{x - x^2}$**, and it is defined only in**

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between $[0, 1]$.

The area enclosed by the curve.

$$\Delta = \int_0^1 (y_1 - y_2)\, dx = 2\int_0^1 \sqrt{x - x^2} dx = 2\int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} dx$$
$$= \left(\frac{2x-1}{4}\sqrt{x - x^2}\Big|_0^1 + \frac{1}{8}\sin^{-1}(2x-1)\Big|_0^1 \right) = \frac{\pi}{4} \text{ sq. units}$$

Q.14 Answer:

$\frac{32}{3}$ sq. units

Solution:

Clearly $y = 2x^2$ **and** $y = x^2 + 4$ **meet when** $2x^2 = x^2 + 4$ **or** $x^2 = 4$

$\therefore x = \pm 2$

Required area $= \left|\int_{-2}^{2} \left(2x^2 - x^2 - 4\right) dx\right| = \left|\int_{-2}^{2} \left(x^2 - 4\right) dx\right| = \left|\left(\frac{x^3}{3} - 4x\right)_{-2}^{2}\right|$

$= \left|\left(\frac{8}{3} - 8\right) - \left| -\frac{8}{3} + 8\right)\right| = \left|\frac{16}{3} - 16\right| = \frac{32}{3}$ sq. Units

Q.15 Answer:

16 sq. unit

Solution:

$\rightarrow$ **representS** $\max(|x|, y\,|) = 4$

$\rightarrow$ **represents** $\min(x|||y|) = 2$

To solve this question we first need to understand the meaning of $\min(|x|, |y|) = 2$

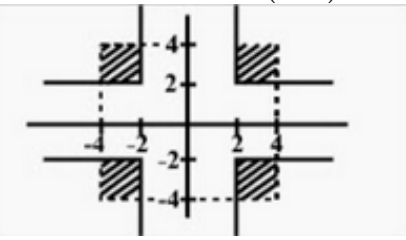
and $\max(|x|, |y|) = 4$

$\min(x), \mid y) = 2$ **means that either** $|x| = 2$ **and** $|y| \geq 2$ **or** $|y| = 2$ **and** $|x| \geq 2$

Similarly, $\max(x), \mid y) = 4$ **means that either** $|x| = 4$ **and** $|y| \leq 4$ **or** $|y| = 4$ **and** $|x| \leq 4$

As can be seen from the image of the graph plotted for this, we can see 4 squares each of size 2×2

So, total area $=4\times(2\times2)=16$



Q.16 Answer:

$a = 3, b = -1$

Solution:

Given curve $y = a\sqrt{x} + bx$

This curve passes through $(1, 2)$; $\therefore 2 = a + b$ **... (i)**

and area bounded by this curve and line $x = 4$ **and** x**-axis is 8 sq. unit**

then $\int_0^4 (a\sqrt{x} + bx)dx = 8$

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$\Rightarrow \quad \frac{2a}{3}\left[x^{3/2}\right]_0^4 + \frac{b}{2}\left[x^2\right]_0^4 = 8, \frac{2a}{3} \cdot 8 + 8b = 8 \quad \Rightarrow \quad 2a + 3b = 3$ **...(ii)**

From equation (i) and (ii), we get $a = 3, b = -1$

Q.17 Answer:

$16 \log 2 + 3$

Solution:

Given the curve, $xy - 3x - 2y - 10 = 0$

$\Rightarrow \mathrm{xy} - 2\mathrm{y} = 3\mathrm{x} + 10$

$\Rightarrow \mathrm{y}(\mathrm{x} - 2) = 3\mathrm{x} + 10$

$\Rightarrow y = \frac{3x+10}{x-2}$

$\therefore$ **Area bounded by the curve** $xy - 3x - 2y - 10 = 0$, x**-axis and the lines** $x = 3, x = 4$

$\int_3^4 |\mathrm{y}| \mathrm{dx}$

$= \int_3^4 \left(\frac{3x+10}{x-2}\right) dx$

$= \int_3^4 \left(\frac{3x-6+16}{x-2}\right) dx$

$= \int_3^4 \left(3 + \frac{16}{\mathrm{x}-2}\right) \mathrm{dx}$

$= [3x + 16 \log |x - 2|]_3^4$

$= [12 + 16 \log |2| - 9 - 16 \log |1|]$

$= 16 \log 2 + 3$ **square units.**

Q.18 Answer:

1 sq. unit

Solution:

Area $= \int_1^{\mathrm{e}} \ln \mathrm{x} \mathrm{dx}$

$= [\mathrm{x} \ln \mathrm{x} - \mathrm{x}|_1^{\mathrm{e}}$

$= (e \ln \mathrm{e} - \mathrm{e}) - (1 \ln 1 - 1)$

$= (e - \mathrm{e}) + 1$

$= 1$

Q.19 Answer:

$c^2 \log \left(\frac{b}{a}\right)$

Solution:

Required area $= \int_a^b \frac{c^2}{x} dx = c^2 |\log x|_a^b = c^2(\log b - \log a) = c^2 \log \frac{b}{a}$

Q.20 Answer:

$\frac{5}{6}$

Solution:

$f'(x) = 2x + 1$

$\Rightarrow f(x) = x^2 + x + c$

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The curve passes through $(1, 2)$, so

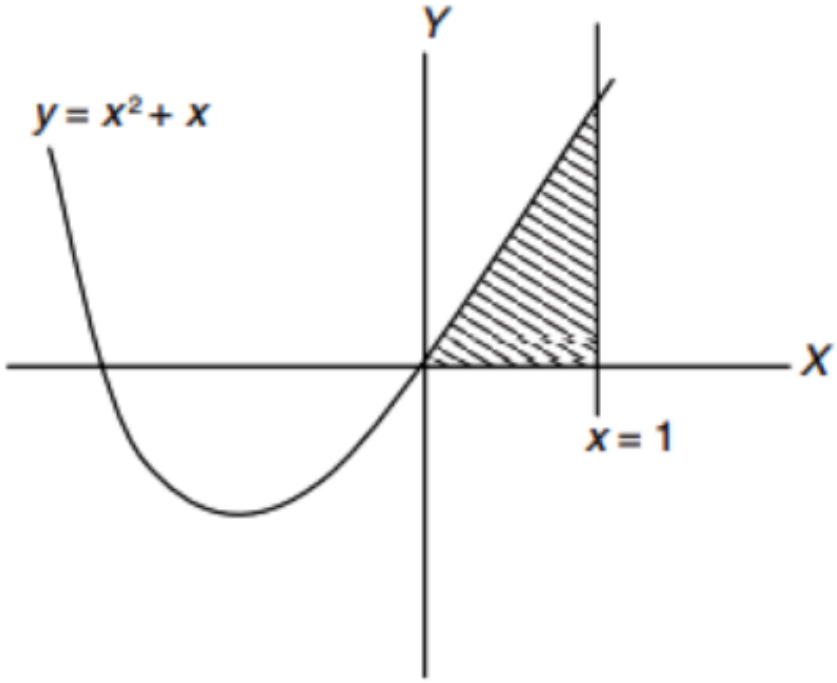
$2 = 1 + 1 + c$

$\Rightarrow c = 0$

$f(x) = x^2 + x$

$\int_0^1 (x^2 + x)\, dx = \left[\frac{x^3}{3} + \frac{x^2}{2}\right]_0^1$

$= \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$ **sq. units**

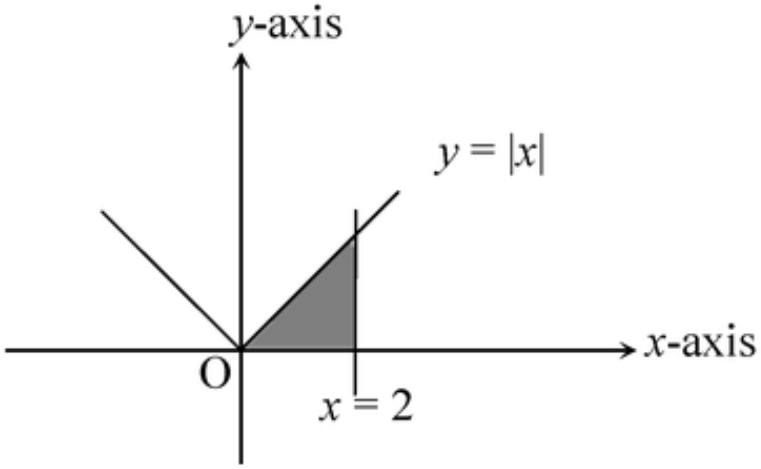


Q.21 Answer:

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Solution:

Area $= 2$sq. **units**



Q.22 Answer:

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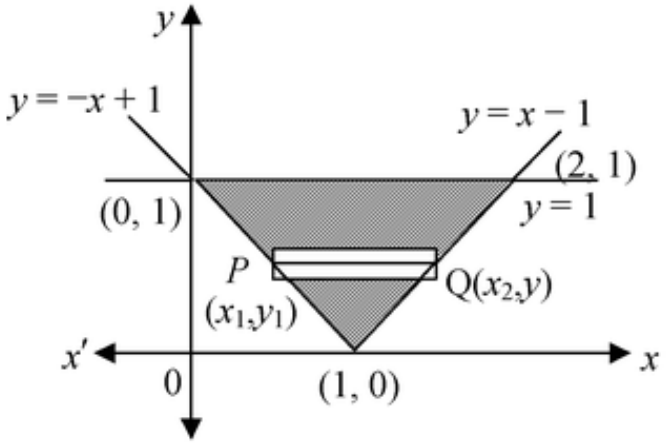
Solution:

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Required area

$= \frac{1}{2}$ base $\times$ height

$= \frac{1}{2} \times 2 \times 1 = 1$



Q.23 Answer:

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Solution:

Required area

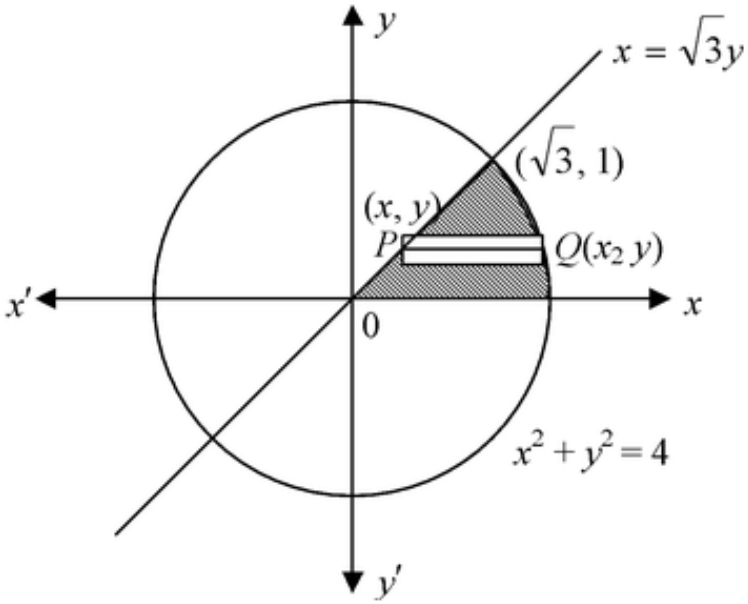
$= \int_0^1 (x_2 - x_1)\, dy$

$= \int_0^1 \left(\sqrt{4 - y^2} - \sqrt{3}y\right) dy$

$= \left[\frac{1}{2}y\sqrt{4 - y^2} + \frac{1}{2}(4)\sin^{-1}\frac{y}{2} - \frac{\sqrt{3}y^2}{2}\right]_0^1$

$= \frac{\sqrt{3}}{2} + 2\sin^{-1}\left(\frac{1}{2}\right) - \frac{\sqrt{3}}{2} - 2\sin^{-1} 0$

$= \frac{\sqrt{3}}{2} + 2\left(\frac{\pi}{6}\right) - \frac{\sqrt{3}}{2} = \frac{\pi}{3}$



Q.24 Answer:

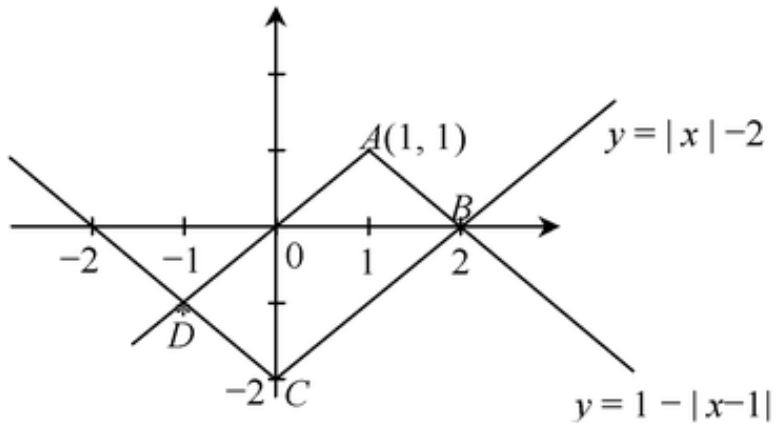
—

Solution:

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Bounded figure $ABCD$ **is a rectangle.** $AB = \sqrt{1+1} = \sqrt{2}; \quad BC = \sqrt{4+4} = 2\sqrt{2}$

Thus, bounded area $= (\sqrt{2})(2\sqrt{2}) = 4$**sq. units**



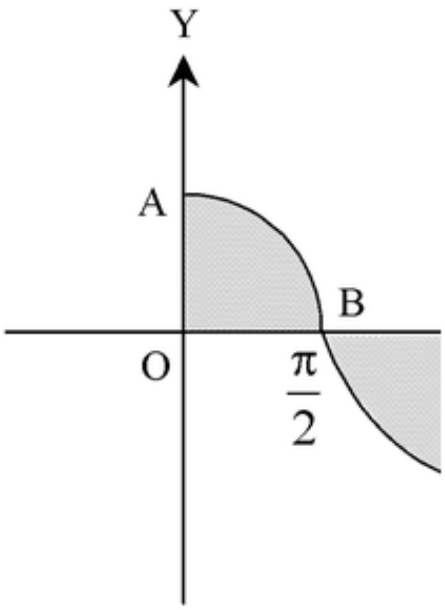
Q.25 Answer:

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Solution:

Required Area $= \left|\int_0^{\pi/2} y dx\right| + \left|\int_{\pi/2}^{3\pi/2} y dx\right| + \left|\int_{3\pi/2}^{2\pi} y dx\right|$

$= 8$ **sq. units**



Q.26 Answer:

—

Solution:

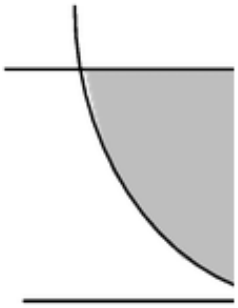
Differentiate both sides

w.r.t. x**, we get**

$\left(e^{2x} - 1\right) f\left(e^x\right) \cdot e^x = \frac{1}{15}\left[15e^{5x} - 15e^{3:}\right.$

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$\left(e^{2x}-1\right) f\left(e^{x}\right) \cdot e^{x}=e^{3x}\left[e^{2x}-1\right]$



$f\left(e^{x}\right)=e^{2x}$
$\Rightarrow f(x)=x^{2}$
Required area $=2\left[8-\int_{0}^{2} x^{2} dx\right]$
square units $=2\left[8-\frac{8}{3}\right]=\frac{32}{3}$

Q.27 Answer:

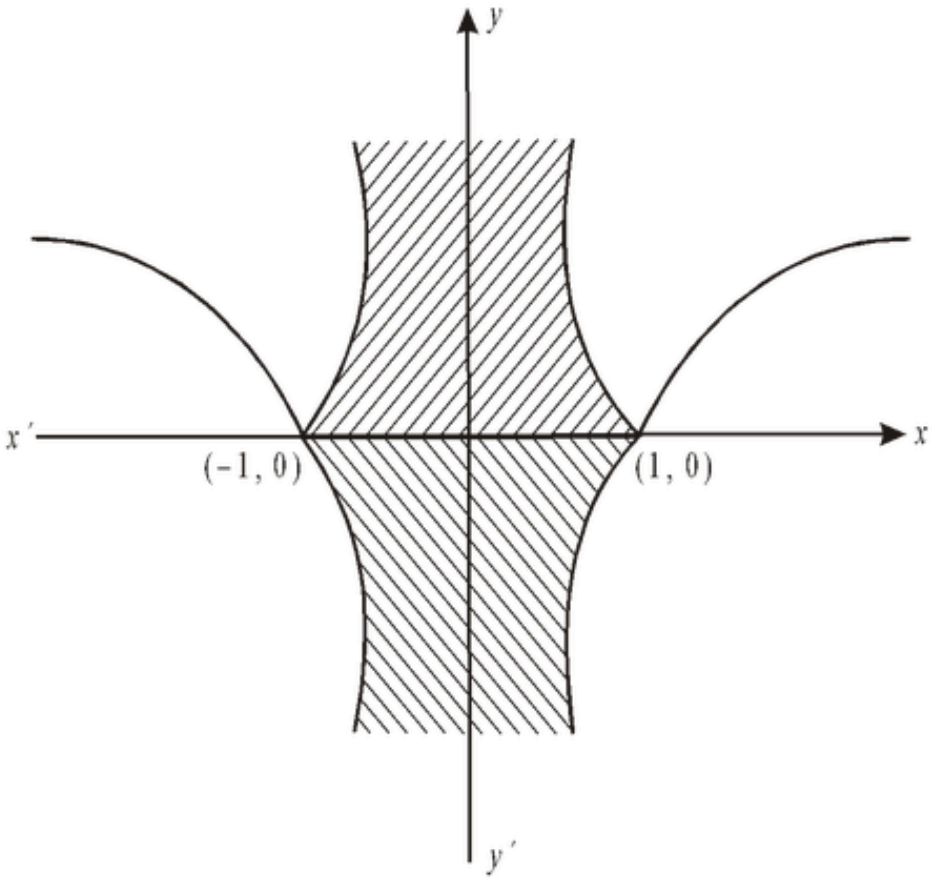
—

Solution:

Required area
= 4
$\int_{0}^{1}|\ln x| dx=-4 \int_{0}^{1} \ln x dx$
$\therefore x \in(0,1), \ln x<0)$

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$= -4 \quad [x \ln x - x]_0^1 = -4$

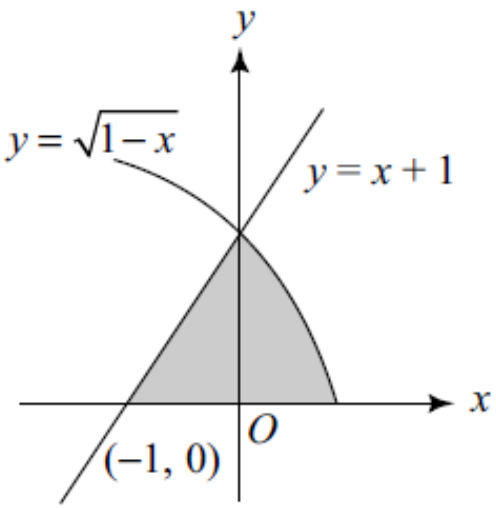


Q.28 Answer:

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Solution:

$$f(x) = \min(x+1, \sqrt{1-x}) = \begin{cases} x+1, & x \le -1 \\ x+1, & -1 < x < 0 \\ \sqrt{1-x}, & 0 \le x \le 1 \end{cases}$$



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$$\text{Required area } = \int_{-1}^{0}(x+1)dx + \int_{0}^{1}\sqrt{1-x}dx$$
$$= \frac{x^2}{2} + x\Big|_{-1}^{0} - \frac{2}{3}(1-x)^{3/2}\Big|_{0}^{1}$$
$$= -\left(-1+\frac{1}{2}\right) + \frac{2}{3} = \frac{7}{6}$$

Q.29 Answer:

—

Solution:

$F(x) = \int_a^x \frac{e^t}{t}dt$, **we need to find**
$g(x) = \int_a^x \frac{e^t}{t+t_0}dt$. **Putting** $t+t_0 = u$ **in** $g(x)$, **we have,**

$$g(x) = \int_{a+t_0}^{x+t_0} \frac{e^{u-t_0}}{u}du = e^{-t_0}\left[\int_a^{x+t_0}\frac{e^u}{u} - \int_a^{a+t_0}\frac{e^u}{u}du\right]$$
$$= e^{-t_0}\left[F\left(x+t_0\right) - F\left(a+t_0\right)\right]$$

Q.30 Answer:

—

Solution:

The area bounded by the curves $y = \frac{x}{6} + \frac{1}{x^2}$, x**-axis,** $x = b$ **and** $x = 2b$ **is given by**

$$f(b) = \int_b^{2b}\left(\frac{x}{6} + \frac{1}{x^2}\right)dx$$
$$= \left(\frac{x^2}{12} - \frac{1}{x}\right)_b^{2b} = \frac{b^2}{4} + \frac{1}{b}$$
$$f'(b) = \frac{2b}{4} - \frac{1}{b^2} = 0 \Rightarrow b = 2^{1/3}$$
$$f''(b) = \frac{1}{2} + \frac{2}{b^3} > 0$$

so $f(b)$ **is minimum for** $b = 1$.

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